

Transaction Practice Questions

Question:

Draw the serializability (precedence) graphs for schedules given below, and state whether each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s).

$S1: r1(X); r2(Z); r1(Z); r3(X); r3(Y); w1(X); w3(Y); r2(Y); w2(Z); w2(Y);$

$S2: r1(X); r2(Z); r3(X); r1(Z); r2(Y); r3(Y); w1(X); w2(Z); w3(Y); w2(Y);$

$S3: r1(X); r3(X); w1(X); r2(X); w3(X);$

Question:

Consider schedules $S3$, $S4$, and $S5$ below. Determine whether each schedule is strict, cascadeless, recoverable, or nonrecoverable. (Determine the strictest recoverability condition that each schedule satisfies.)

$S3: r1(X); r2(Z); r1(Z); r3(X); r3(Y); w1(X); c1; w3(Y); c3; r2(Y); w2(Z); w2(Y); c2;$

$S4: r1(X); r2(Z); r1(Z); r3(X); r3(Y); w1(X); w3(Y); r2(Y); w2(Z); w2(Y); c1; c2; c3;$

$S5: r1(X); r2(Z); r3(X); r1(Z); r2(Y); r3(Y); w1(X); c1; w2(Z); w3(Y); w2(Y); c3; c2;$

Question:

How many *serial* schedules exist for the three transactions in Figure given below? What are they? What is the total number of possible schedules?

Transaction T_1	Transaction T_2	Transaction T_3
<code>read_item(X);</code>	<code>read_item(Z);</code>	<code>read_item(Y);</code>
<code>write_item(X);</code>	<code>read_item(Y);</code>	<code>read_item(Z);</code>
<code>read_item(Y);</code>	<code>write_item(Y);</code>	<code>write_item(Y);</code>
<code>write_item(Y);</code>	<code>read_item(X);</code>	<code>write_item(Z);</code>
	<code>write_item(X);</code>	

Question:

Why is an explicit transaction end statement needed in SQL but not an explicit begin statement?

Question:

Prove that the basic two-phase locking protocol guarantees conflict serializability of schedules. (*Hint*: Show that if a serializability graph for a schedule has a cycle, then at least one of the transactions participating in the schedule does not obey the two-phase locking protocol.)

Question:

What is the two-phase locking protocol? How does it guarantee serializability?

Question:

What are some variations of the two-phase locking protocol? Why is strict or rigorous two-phase locking often preferred?

Question:

Describe the wait-die and wound-wait protocols for deadlock prevention.